BRAC University (Department of Computer Science and Engineering)

CSE 221 (Algorithm) for Spring 2025 Semester

Quiz 1 (Set A)

Student ID:
Section: **Full Marks: 20**
Name: **Duration: 30 minutes**

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1. Given a program, write down the time complexity. 5

```
int i, j, k, a, b, sum
for ( i = 0; i < n; i = i + 3)
      for ( j = n; j >= 1; j = j / 5)
            for ( k = 1; k <= n; k = k * 5)
                  sum = a + b
```

2. Given a program, write down the time complexity. 4

```
for i in range(1, n):
    j = 1
    while j < i*i:
        j = j + 1
```

3. You are given a sorted binary array **arr** of length n, where each element is either **0** or **1**. The array is sorted in **non-decreasing** order, meaning that all 0s appear before any 1s. Your task is to find the **number of 1s** in the array in less than O(N) time complexity.

Example: arr = [0, 0, 0, 1, 1, 1, 1] Output: 4

Propose an algorithm for this task. 4

4. You have recently joined Facebark as a software engineer. It's a new social media platform where people share pictures of their dogs. There is a newsfeed where posts by Facebark friends appear. Each user has a numerical ID. You are trying to figure out the order of the newsfeed posts. There have been numerous complaints about users who spam the platform by posting a thousand pictures simultaneously. Therefore, you came up with an idea: the posts by users will be shown in ascending order of their post counts. So, you printed the data of the counts of the newest posts:

id	0	1	2	3	4	5	6	7	8	9
post count	8	12	3	5	2	17	9	14	2	3

After looking at the data, you thought, “Oh, easy! I can write a linear time algorithm for this!”

a. Describe the algorithm by showing the steps of how you would sort the above list in linear time. 5

Two days after this implementation, the entire server crashed after running out of memory. You found out that this happened after a few users uploaded 1000 pictures each. But people uploaded thousands of pictures regularly before and this never happened.

b. Do you think the crash was a result of your sorting algorithm? If the answer is yes, provide a possible explanation about how this could have happened. 2

BRAC University (Department of Computer Science and Engineering)

CSE 221 (Algorithm) for Spring 2025 Semester

Quiz 1 (Set B)

Student ID:
Section: **Full Marks: 20**
Name: **Duration: 25 minutes**

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1. Given a program, write down the time complexity 5

```
int i, j, k, m, a, b, sum
for (i=0; i<n; i+=4) {
    for (j=1; j<n; j*=2) {
        for (k=0; k<30; k++) {
            sum += a
        }
        print("Done");
        for (m=n; m>0; m-=2) {
            Sum += b
        }
    }
}
```

2. Given a program, write down the time complexity. 4

```
for (i = n / 2; i > 1; i /= 6) {
    for (j = 2; j <= i; j *= 4) {
        for (k = 0; k <= j; k *= 3) {
            p = p + n / 2;
        }
    }
}
```

3. You are given a sorted binary array **arr** of length n, where each element is either **0** or **1**. The array is sorted in **non-decreasing** order, meaning that all 0s appear before any 1s. Your task is to find the **number of 1s** in the array in less than O(N) time complexity.

Example: arr = [0, 0, 0, 1, 1, 1, 1] Output: 4

Propose an algorithm for this task. 4

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b. Do you think the crash was a result of your sorting algorithm? If the answer is yes, provide a possible explanation about how this could have happened. 2

BRAC University (Department of Computer Science and Engineering)
CSE 221 (Algorithms) for Spring 2025 Semester
Quiz 2
Set A

Student ID:
Section: **Full Marks: 20**
Name: **Duration: 30 minutes**

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1. Find the time complexity of the following recurrence relation:
[Any method is acceptable as long as steps are shown]

$$T(n) = T(n/2) + T(n/4) + n$$ 5

2. 4. Find the time complexity of the following recurrence relation: [Use Master Theorem]

$$T(n) = 6T(n/3) + \frac{n^2}{logn}$$ 4

3. Take a look at the following recursive algorithm which finds the value of xn:

```
def power(x, n):
    if n == 0:
        return 1

    pow = power(x, n//2)

    # n is odd
    if n % 2 == 1:
        return x * pow * pow

    # otherwise n is even
    return pow * pow
```

Find the recurrence relation of the given algorithm as a function of T(n). Also find the runtime of the algorithm using that function. 4

4. Imagine yourself helping Taylor Swift following her "Eras Tour first show." Taylor is curious about her performance and the groups of songs her audience connected most with them. She receives comments for every song on her setlist: some calm the audience down a bit (***and have a negative score***), while others get the crowd quite thrilled (***and have a positive score***).

Taylor can choose any song to start her performance, but she likes to keep the flow in order. So the next song she sings has to be the one right after she has chosen the starting song on the list, regardless of the public reactions**.**
Given a list of scores for each song (indicating how much they boost or calm the crowd), can you:

[-3, 7, 12, -8, -2, 83, -7, 4]

i. Help Taylor determine which segment of consecutive songs she should perform at her next concerts to maximize audience interest. What is the total comment score of this segment? **Simulate** with proper steps. 5

ii. **Explain** the time taken by your preferred method, particularly given the *"n"* numbers of songs Taylor has in her setlist. Ideally, your method should not exceed $O(nlogn)$. 2

BRAC University (Department of Computer Science and Engineering)
CSE 221 (Algorithms) for Spring 2025 Semester
Quiz 2
Set B

Student ID:
Section: **Full Marks: 20**
Name: **Duration: 30 minutes**

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1. Find the time complexity of the following recurrence relation:
[Any method is acceptable as long as steps are shown]

$$T(n) = T(n/3) + T(n/5) + n$$ 5

2. Find the time complexity of the following recurrence relation: [Use Master Theorem]

$$T(n) = 64T(n/8) + n^2 logn$$ 4

3. Take a look at the following recursive algorithm which finds the value of xn:

```
def power(x, n):
    if n == 0:
        return 1

    pow = power(x, n//2)

    # n is odd
    if n % 2 == 1:
        return x * pow * pow

    # otherwise n is even
    return pow * pow
```

Find the recurrence relation of the given algorithm as a function of T(n). Also find the runtime of the algorithm using that function. 4

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Taylor can choose any song to start her performance, but she likes to keep the flow in order. So the next song she sings has to be the one right after she has chosen the starting song on the list, regardless of the public reactions**.**

Given a list of scores for each song (indicating how much they boost or calm the crowd), can you:

a) **[-3, 7, 12, -8, -2, 83, -7, 4]**

i. Help Taylor determine which segment of consecutive songs she should perform at her next concerts to maximize audience interest. What is the total comment score of this segment?

Simulate with proper steps. 5

ii. **Explain** the time taken by your preferred method, particularly given the *"n"* numbers of songs Taylor has in her setlist. Ideally, your method should not exceed $O(nlogn)$. 2

BRAC University (Department of Computer Science and Engineering)

CSE 221 (Algorithms) for Spring 2024 Semester

Quiz 3

Set A

Student ID:

Section: **Full Marks: 20**

Name: **Duration: 30 minutes**

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1. Suppose we have the following social network, Friendsbook where:

Person A is friends with Person B, C, D, and E.
Person B is friends with Person A, C, and F.
Person C is friends with Person A, B, D, and G.
Person D is friends with Person A, C, E, G, and H.
Person E is friends with Person A, D, and H.
Person F is friends with Person B and G.
Person G is friends with Person C, D, F, and H.
Person H is friends with Person D, E, and G.

A. Draw a graph to represent the network using each user as a node or vertex and the friendships as edges between the nodes. 2
B. Create the adjacency list and adjacency matrix for the graph representation. 2
C. For each pair of users, calculate the number of friends they have in mutual. 2
D. In Friendsbook, a user can share posts. Suppose the CEO of Friendsbook hired you for developing their post display system. You were given the task to implement a system that displays posts from users who are at most two degrees of connection away from the one who is scrolling his feed. After looking at the requirements you came up with an idea to implement the system using a particular algorithm. Now, for person F, simulate your algorithm. Can F see the posts of all other users in his feed? 4

2. Your friend Habib said that all the countries in the world can be colored with only 2 colors where no 2 adjacent (if 2 countries share borders that means they are adjacent) countries have the same colors. You wanted to check whether Habib was correct or not so you took a particular portion of the world map which is described below.

India has border with Nepal, Pakistan, Bangladesh and Myanmar
China has border India, Pakistan, Nepal, Bhutan and Myanmar
Pakistan has border with India and China
Bangladesh has border with India and Myanmar
Myanmar has border with Bangladesh,India and China
Nepal has border with India and China
Bhutan has border with India and China

a) Draw a graph where each country will represent a Vertex and if they share a border they will have an edge between them. 2
b) Mention a graph algorithm which you are gonna use to find out whether a graph is a 2-colorable or not and also write a pseudocode/written explanation of your procedure. 4
c) Simulate the graph in (a) using your pseudo code to find out whether Habib was correct or not. 4

BRAC University (Department of Computer Science and Engineering)

CSE 221 (Algorithm) for Spring 2025 Semester

Quiz 4

Set A

Student ID:

Section: **Full Marks: 20**

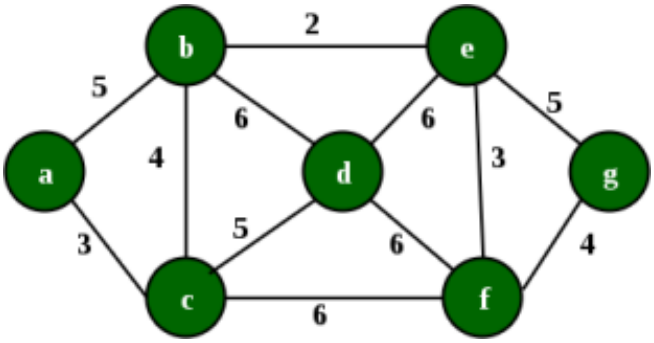
Name: **Duration: 20 minutes**

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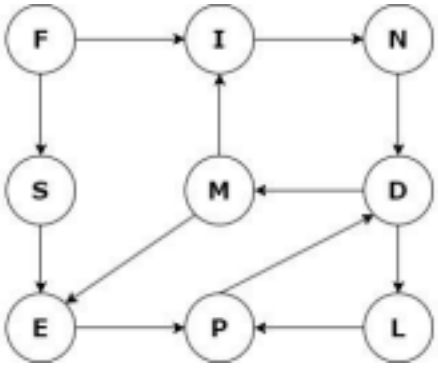
1. You work as an engineer for the Roads and Highways department in a district. The district has 7 thanas that are represented by the 7 vertices in the following graph. The edges between the vertices represent the roads that connect one thana with another and the weight of an edge represents the length of the road.

A recent flood has totally damaged these roads and immediate repair work is needed. The cost to repair a road is proportionate to its length. However, your department does not have the budget to repair all the roads so you need to repair a subset of the roads. Which algorithm should you use here to find out the roads that need to be repaired to keep all the thanas connected to one another with the **minimum** possible cost?

Show the step by step simulation of the algorithm. You may pick any algorithm of your preference that you think will solve the problem. 10



2. Sherlock Holmes found himself in a dire situation where James Moriarty imprisoned his assistant Watson in a maze. The maze is made of 9 different rooms.



When Sherlock tries to save his friend, water suddenly fills the rooms. He needs to find a way to contain the water somehow so that the water doesn't overflow from one place to another. So, he created different isolated closed groups of rooms where he can travel anywhere in that particular group, meaning they have a path from any room to any other room inside the group. A group can contain 1 to N number of rooms where N is the total number of vertices, and no two groups can share the same room.

i) **Mention** the name of a suitable algorithm for this task. 2

ii) **Show** simulation for the graph given above. 8